REINFORCEMENT LEARNING

A SIMPLE EXAMPLE: Balancing a Pole on a Cart (CartPole)

REINFORCEMENT LEARNING

The Actor-Critic Method

Goal: learn the **best action to take in each situation** by combining two cooperating networks.

THE ACTOR

Decides what to do

A policy network that, for the current state, outputs a probability over possible actions and picks one.

“proposes the move”

THE CRITIC

Judges how good it was

A value network that estimates the expected future reward of the state, giving feedback on the Actor’s choice.

“scores the move”

Why use it?

- Works with huge / continuous state spaces (uses neural networks).
- Learns step-by-step (Temporal Difference Learning). No need to wait for the episode to end.
- Critic’s feedback lowers variance, so training is more stable & faster.

A SIMPLE EXAMPLE:

Balancing a Pole on a Cart (CartPole)

A cart moves left or right on a track; a pole is hinged on top. The agent must keep the pole upright.

The learning loop

STATE: cart position, velocity, pole angle, pole velocity

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ACTOR → push LEFT or RIGHT

↓

REWARD +1 for every step the pole stays up

↓

CRITIC → was that better or worse than expected? (advantage)

Who does what here

Actor: Sees the pole tilting right → raises the probability of pushing right.

Critic: Predicts how long the pole can stay up from this state, and grades the action.

Update: If the action beat the Critic's expectation, make it more likely next time.

RESULT

After repeating this loop, the agent reliably keeps the pole balanced. The same recipe scales up to robots, game AI, and self-driving control.

Example based on the TensorFlow CartPole-v0 Actor-Critic tutorial.

FROM TOY PROBLEM TO INDUSTRY

Where Actor-Critic Leads

The same state → action → reward loop now powers some of the most valuable systems in tech. PPO, today's standard large-scale variant, is a direct descendant of this method.

GENERATIVE AI

LLM Alignment (RLHF)

Action
the model's next reply

Reward
human preference score

PPO is an Actor-Critic algorithm that became the standard method for tuning assistants such as ChatGPT and Claude. Newer options like DPO/GRPO now exist too.

AD TECH

Real-Time Bidding

Action
how much to bid for an impression

Reward
click / conversion value

Reinforcement learning is increasingly applied here to set bids in real time, aiming to maximize return on ad spend across high-volume auctions.

REVENUE / RETAIL

Dynamic Pricing

Action
the price to set right now

Reward
profit, demand, inventory

Reinforcement learning is being explored to adjust prices in response to demand and competition, with applications in travel, ride-hailing and e-commerce.

Reality check: Reinforcement learning shines in a few high-impact, well-defined domains. It is powerful, not a cure-all. Reward design and stable deployment remain the hard parts.